Min Jae Bae

Ph.D. Candidate · Plant Epigenetics and Chromatin Biology

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EDUCATION

Ph.D. candidate in Biomedical Science and Engineering, Konkuk University Mar. 2021 – present (expected Feb. 2027)
Advisor: Prof. Dae-Jin Yun. Research: HOS15-mediated degradation of linker histone H1.3 and LisH-dependent HOS15 dimerization in cold stress.

M.S. in Biomedical Science and Engineering, Konkuk University Mar. 2019 – Feb. 2021
Thesis: Structural and Functional Analysis of HOS15, a Substrate Receptor of CRL4 in Cold Stress Signaling. Advisor: Prof. Dae-Jin Yun.

B.S. in Biomedical Science and Engineering, Konkuk University Mar. 2013 – Feb. 2019
Thesis: Functional Studies of SUMOylation of Histone Deacetylase in Plants. Advisor: Prof. Dae-Jin Yun.

RESEARCH FOCUS

My research asks how plants remodel chromatin to survive cold. I study HOS15, the WD40-repeat substrate receptor of the CULLIN4–DDB1 E3 ubiquitin ligase ($CRL4^{HOS15}$), and its chromatin substrates, the histone deacetylase HD2C and the linker histone H1.3, in *Arabidopsis thaliana*.

Linker histone H1.3 and chromatin decondensation in the cold

I found that HOS15 interacts preferentially with H1.3 among the three Arabidopsis H1 variants, that cold lowers H1 protein in a HOS15- and proteasome-dependent manner, and that H1.3 is the variant that is lost. *h1.3* mutants are more freezing tolerant and suppress the freezing sensitivity of *hos15-2*. Their chromocenters are fewer and less condensed, resembling cold-treated wild-type nuclei, and NicE-seq shows increased accessibility mainly in pericentromeric heterochromatin. RNA-seq shows that the core CBF–COR program is intact in *h1.3*, whereas defense- and lignin-related genes are pre-activated and heat-shock and antioxidant genes are induced by cold only in *h1.3*.

HOS15 dimerization and $CRL4^{HOS15}$ activity

I showed that HOS15 forms homodimers through its N-terminal LisH motif. Alanine substitutions of conserved hydrophobic LisH residues abolish dimerization in yeast, and these variants fail to support cold-induced ubiquitination and degradation of HD2C or to restore the freezing tolerance of *hos15-2*. I also characterize the intrinsically disordered region (IDR) within the HOS15 coiled-coil domain, and use LC-MS-based proteomics to profile proteins that change in *hos15* and *h1* mutants.

RESEARCH PROJECTS

- **Global Plant Stress Research Center** (NRF), 2024.08–2030.04 — Participating Researcher
- **Genome-wide chromatin reprogramming and transcriptome regulation in plant environmental stress adaptation** (NRF), 2026.03–2031.02 — Participating Researcher
- **Chromatin reconfiguration and transcriptional control for plant stress adaptation** (NRF), 2022.03–2027.02 — Co-participating Researcher

PROFESSIONAL SERVICE

- **Member, NASA GeneLab OSDR Plant Analysis Working Group (Plant AWG)**, Sep. 2026–present — Contributing to the analysis of spaceflight and astrobotany omics datasets in NASA's Open Science Data Repository (OSDR), with international researchers on plant biology in the space environment.

- **Member, NASA GeneLab OSDR Regolith Analysis Working Group (Regolith AWG)**, Sep. 2026–present — Contributing plant molecular biology expertise to cross-disciplinary work on plant and plant–microbe growth in lunar/Martian regolith, including dataset curation and standards in NASA OSDR.

SKILLS

- **Plant molecular biology:** Gateway cloning, Gibson assembly, site-directed mutagenesis, vector construction and design (Vector NTI), transgenic line generation, Arabidopsis genetics and phenotypic analysis
- **Epigenetics & chromatin biology:** histone biology, chromatin remodeling, histone acetylation/methylation analysis, nucleosome organization, NicE-seq, ChIP-seq, open-chromatin/accessibility analysis
- **Protein structure & function:** domain-level mutagenesis, intrinsically disordered region (IDR) analysis, protein conformation and structure–function studies
- **Omics & data analysis:** RNA-seq and NicE-seq analysis (R: DESeq2, limma, GSEA; SalmonProcessor, OmicsForge), LC-MS proteomics, alternative splicing analysis, sequence alignment (BioEdit), genome data visualization (IGV)
- **Protein & stress biology:** ubiquitin–proteasome pathway, E3 ligase complexes, protein stability, cold/drought/thermotolerance and senescence signaling
- **Software development:** PhytoOmics Studio, an R/Shiny application for plant multi-omics analysis (in development; RNA-seq module with DESeq2, GO/KEGG enrichment and GSEA for four plant species; Windows installer with bundled R)
- **Software/tools:** R, HOMER, IGV, BioEdit, Vector NTI, GraphPad Prism, Bio-Rad CFX Maestro (RT-qPCR), Zeiss ZEN and Leica LAS X (confocal imaging), ImageJ

PUBLICATIONS (SCIE/ESCI)

1. Ali A, Zareen S, Bader ZE, Park J, Khan IU, Park K, Albakri N, **Bae MJ**, Bressan RA, Pardo JM, Yun DJ, Xu ZY. A GPS2-like protein interacts with HOS15 and HDA6 to form a repressor complex that regulates ABA signaling and drought adaptation in Arabidopsis. *Plant Communications* (2026). doi:10.1016/j.xplc.2026.101843
2. **Bae MJ**, Zareen S, Yun H, Ali A, Bader ZE, Park K, Albakri N, Jung YJ, Yun DJ, Park J. Unveiling the epigenetic and stress-responsive function of histone H1 in plants. *Plant Signaling & Behavior* (2026). doi:10.1080/15592324.2026.2677313
3. Zareen S, Ali A, **Bae MJ**, Albakri N, Park K, Yun H, Yun DJ, Park J. Arabidopsis E3 ligase HOS15 regulates ABA and drought responses via DIL9 degradation. *Journal of Plant Growth Regulation* (2026). doi:10.1007/s00344-026-12240-w
4. Albakri N, Bader ZE, Park KS, **Bae MJ**, Zareen S, Yun H, Pradhan S, Yun DJ, Park J. HOS15 modulates alternative splicing homeostasis under cold stress in *Arabidopsis thaliana*. *Journal of Plant Biology* (2025). doi:10.1007/s12374-025-09465-9
5. Park K, Lim CJ, Albakri N, **Bae MJ**, Park J. HOS15 contributes to thermotolerance through destabilization of HD2C in Arabidopsis. *Journal of Plant Biology* (2025). doi:10.1007/s12374-025-09466-8
6. Zareen S, Ali A, **Bae MJ**, Albakri N, Park K, Yun H, Yun DJ, Park J. HOS15 together with PWR–HDA9 positively regulates dark-induced senescence in Arabidopsis. *Plant Signaling & Behavior* (2025). doi:10.1080/15592324.2025.2564962
7. Bader ZE[†], **Bae MJ**[†], Ali A, Park J, Baek D, Yun DJ. GIGANTEA-ENHANCED EM LEVEL complex initiates drought escape response via dual function of ABA synthesis and flowering promotion. *Plant Signaling & Behavior* (2023). doi:10.1080/15592324.2023.2180056
8. Lim CJ, Park KS, Ali A, Park J, Ryou SM, Shen M, Khan HA, Bader ZE, Zareen S, **Bae MJ**, Choi JH, Xu ZY, Pardo JM, Yun DJ. Negative regulation of floral transition in Arabidopsis by HOS15-PWR-HDA9 complex. *Frontiers in Plant Science* (2023). doi:10.3389/fpls.2022.1105988
9. Lim CJ, Park J, Shen M, Park HJ, Cheong MS, Park K, Baek D, **Bae MJ**, Ali A, Jan M, Lee SY, Lee BH, Kim WY, Pardo JM, Yun DJ. The histone-modifying complex PWR/HOS15/HD2C epigenetically regulates cold tolerance. *Plant Physiology* (2020). doi:10.1104/pp.20.00439

[†] These authors contributed equally.

Manuscripts in preparation

- **Bae MJ**, Bader ZE, Ali A, Yun DJ, Park J. Mechanisms of histone H1 degradation under cold stress.
- Ali A[†], **Bae MJ[†]**, Bader ZE, Zareen S, Park B, Ali Z, Yun DJ, Park J. HOS15 dimerization promotes HD2C degradation during cold stress.

PATENTS

- Vector for improving productivity of cellulose and transformed mouseear cress by the same. KR Patent Application No. 10-2023-0113597
- Vector for improving productivity of cellulose and transformed mouseear cress by the same. KR Patent Application No. 10-2024-0168432
- Vector for improving productivity and storage capacity of vegetable oil and transformed plant by the same. KR Patent Application No. 10-2024-0168464
- Ethanol production and tolerance enhanced yeast strain and method for producing ethanol using the same. KR Patent Application No. 10-2025-0104960
- Sucrose-accumulating plant and method for preparing the same. KR Patent Application No. 10-2025-0104961
- Recombinant *Kluyveromyces marxianus* strain with engineered xylose metabolic pathway and method for producing ethanol using the same. KR Patent Application No. 10-2025-0157597

AWARDS AND HONORS

HOPE Fellow, HOPE Meetings with Nobel Laureates, Japan Society for the Promotion of Science (JSPS), Japan — Mar. 2024

Excellent Poster Award, Korean Society of Plant Biologists (KSPB), Korea — Nov. 2021

CONFERENCES AND PRESENTATIONS

- 2026 GPSRC International Conference, Konkuk University, Korea, May 2026
- 2026 Winter Conference of the Korean Society of Plant Biologists, Korea University, Korea, Feb. 2026 — Poster
- 2nd Global Plant Stress Research Center (GPSRC) & 2025 KSMCB Plant Molecular Biology International Symposium, Seoul, Korea, Aug. 2025 — Organizer
- 2024 International Conference of the Korean Society of Plant Biologists, Seogwipo, Korea, Nov. 2024 — Oral presentation
- 15th HOPE Meeting, Kyoto, Japan, Feb. 2024 — Poster
- 2022 International Conference of the Korean Society of Plant Biologists, Busan, Korea, Nov. 2022 — Poster
- 2021 International Conference of the Korean Society of Plant Biologists, online, Nov. 2021 — Poster
- 2020 International Conference of the Korean Society of Plant Biologists, online, Nov. 2020 — Poster
- 2019 Annual Conference of the Korean Society of Plant Biologists, Buyeo, Korea, Nov. 2019 — Poster
- 2018 International Conference of the Korean Society of Plant Biologists, Busan, Korea, Nov. 2018 — Poster